

ADDITIONAL SIMULATIONS

Tijn Jacobs

June 4, 2026

This file contains some specific simulations for the Supplementary Materials. They should answer specific questions, or investigate properties of specific parts of the model.

1 Simulation on the nonparametric error distribution

1.1 Goal

The methodology section introduces a centered hierarchical Dirichlet process (HDP) prior on the residual distribution, with the option of a source-specific scale parameter σ_s . This simulation has three aims.

1. **Demonstrate the practical benefit of the HDP $_{\sigma}$ prior.** Under data-generating processes where the two sources share residual shape but differ in scale—and especially when one source is too small to fit a flexible nonparametric residual on its own—atom sharing should yield lower CATE estimation error, narrower credible intervals without sacrificing coverage, and a more accurate per-source residual distribution than each of the five alternative residual priors.
2. **Map out where the various priors do and do not differ.** The DGPs are designed so that each successive scenario breaks an assumption that one of the simpler priors relies on: Gaussianity, equality of source scales, equality of source shapes, and finally rarity of a residual “type” in one source but not the other. We expect a clean staircase of relative performance, with HDP $_{\sigma}$ either tied for best or strictly best in every cell.
3. **Probe the identifiability of the hierarchical parameters.** The atoms θ_k^* and the per-source centering shifts μ_s are not separately identified—only the centered locations $\theta_k^* - \mu_s$ are. We track posterior summaries of $|\mu_0 - \mu_1|$ and of the maximum unconstrained atom magnitude $\max_k |\theta_k^*|$ to confirm that this tension affects component-level interpretation without degrading the CATE estimates that the model targets.

1.2 Estimands and metrics

For each replicate we record the following summaries.

CATE estimation. Posterior-mean root mean squared error $\text{RMSE}_{\tau} = (n^{-1} \sum_i (\hat{\tau}(X_i) - \tau(X_i))^2)^{1/2}$, bias $\text{Bias}_{\tau} = n^{-1} \sum_i (\hat{\tau}(X_i) - \tau(X_i))$, nominal-95% credible-interval coverage aver-

aged over i , and mean credible-interval width. CATE estimates are evaluated on the training sample with $\tau(x) = 1.5 + 2x_1$.

Per-source residual scale recovery. We report two summaries.

- *Within-component scale.* The posterior mean of the within-component standard deviation σ (or σ_s in the scale modes). This is directly comparable to the truth only when the truth is Gaussian; under mixture truths the within-component σ is intentionally smaller than the total residual sd, so cross-mode comparison on this metric is misleading.
- *Total residual standard deviation.* The posterior mean of

$$\hat{\sigma}_s^{\text{tot}} = \sqrt{\hat{\sigma}_s^2 + \sum_k \hat{\pi}_{sk} (\hat{\theta}_{sk} - \hat{\mu}_s)^2}$$

compared against the empirical standard deviation of the simulated source-specific residuals. This is the apples-to-apples metric across all six priors.

Per-source residual shape recovery. Two-sample Kolmogorov–Smirnov statistic between the empirical CDF of the posterior-mean residuals $\{Y_i - \hat{Y}_i : S_i = s\}$ and the true residuals $\{\varepsilon_i : S_i = s\}$ drawn during data generation. Low values indicate that the prior is recovering the source-specific residual law.

Mixture diagnostics.

- *Effective number of components* per source, $K_s^{\text{eff}} = 1/\sum_k \pi_{sk}^2$, averaged over MCMC iterations; the discrete analogue is the number of components with $\pi_{sk} > 0.01$.
- *Atom sharing:* the number of components with both $\pi_{0k} > 0.01$ and $\pi_{1k} > 0.01$ (per iteration, averaged). Only meaningful for the source-stratified priors (iDP, iDP $_{\sigma}$, HDP, HDP $_{\sigma}$).
- *Concentration parameters:* posterior means of the per-source concentration M_s and, for the HDP priors, the top-level concentration γ .

HDP identifiability probes. For the HDP priors we additionally record the posterior means of $|\mu_0 - \mu_1|$ and of $\max_k |\theta_k^*|$. Both quantities should remain on the same scale as the true mean shift implied by the DGP and the atom-prior scale σ_{θ} , respectively; large values flag the $\mu_s - \theta_k^*$ identifiability tension noted in Section ??.

1.3 Compared priors

We compare six residual priors. They share a common skeleton and differ along three axes:

1. **Shape:** parametric Gaussian vs. nonparametric Dirichlet-process (DP) mixture of Gaussians;
2. **Stratification:** residual distribution pooled across sources vs. stratified by source;

3. **Borrowing:** when stratified, whether information is borrowed across sources via a hierarchical DP (HDP);

with an orthogonal fourth axis for the residual scale:

4. **Scale:** a single residual scale σ shared by both sources vs. source-specific scales σ_0, σ_1 .

We first set up the common building block, then state each prior as a single small modification of the previous one.

Common building block: centered DP mixture of Gaussians. A Dirichlet-process mixture of Gaussians (?) models a residual as a mixture

$$f(w | G, \sigma) = \int \frac{1}{\sigma} \phi\left(\frac{w - \theta}{\sigma}\right) dG(\theta), \quad G \sim \text{DP}(\alpha, H),$$

with ϕ the standard normal density and base measure $H = N(0, \sigma_\theta^2)$. For computation we use the Ishwaran–James truncation (?) at K components, so $G = \sum_{k=1}^K \pi_k \delta_{\theta_k}$ with $\pi \sim \text{Stick}_K(\alpha)$ and $\theta_k \stackrel{\text{iid}}{\sim} N(0, \sigma_\theta^2)$. Mean-zero centering (?) replaces the atoms θ_k by $\theta_k - \mu$ with $\mu = \sum_k \pi_k \theta_k$, ensuring $\mathbb{E}[\varepsilon] = 0$ so that the structural-mean components m_0, τ, c retain their interpretation.

1. **G(baseline).** A single Gaussian residual, shared across sources:

$$\varepsilon_i | \sigma \sim N(0, \sigma^2).$$

σ has the usual scaled-inverse- χ^2 prior.

2. **sDP(one DP, pooled across sources).** Replace the Gaussian by a single centered DP mixture, pooled across both sources:

$$\varepsilon_i | G, \sigma \sim f(\cdot | G, \sigma) \quad \text{for all } i,$$

with G as in the common building block. Residual shape is allowed to be non-Gaussian but is assumed identical across sources.

3. **iDP(per-source DP, independent).** Fit a separate centered DP mixture per source, with no cross-source borrowing. Two independent mixing measures G_0, G_1 , common σ :

$$\varepsilon_i | S_i = s, G_s, \sigma \sim f(\cdot | G_s, \sigma), \quad G_s = \sum_{k=1}^K \pi_{sk} \delta_{\theta_{sk}} \text{ independently across } s.$$

Sources can now differ in residual *shape*.

4. **iDP $_\sigma$ (per-source DP + per-source scale).** Identical to iDP but with source-specific scales:

$$\varepsilon_i | S_i = s, G_s, \sigma_s \sim f(\cdot | G_s, \sigma_s),$$

with independent scaled-inverse- χ^2 priors on each σ_s^2 .

Table 1: The six residual priors compared in this simulation, arranged by residual-shape construction (rows) and scale parameterisation (columns). For each prior we give the math symbol (used throughout this document) and the R argument value (used in the `FusionForest()` call).

Residual shape	Shared σ		Per-source σ_s	
	symbol	R argument	symbol	R argument
Gaussian	G	<code>gaussian</code>	—	—
Pooled DP mixture	sDP	<code>shared_dp</code>	—	—
Per-source DP mixtures, independent	iDP	<code>source_dp</code>	iDP_σ	<code>source_dp_scale</code>
Per-source DP mixtures with HDP atom-sharing	HDP	<code>source_hdp</code>	HDP_σ	<code>source_hdp_scale</code>

5. HDP(HDP, shared σ). The methodology section’s prior. Replace the independent G_0, G_1 of iDP by a hierarchical DP (?) that ties them through a top-level discrete measure:

$$G^* \sim \text{DP}(\gamma, H), \quad G_s^* \sim \text{DP}(M_s, G^*) \quad \text{for } s \in \{0, 1\}.$$

Equivalently, after truncation, $G^* = \sum_k \beta_k \delta_{\theta_k^*}$ has *shared atoms* θ_k^* , and each per-source measure $G_s^* = \sum_k \pi_{sk} \delta_{\theta_k^*}$ shares those atoms but re-weights them by source-specific π_{sk} . After centering,

$$\varepsilon_i | S_i = s, G_s^*, \sigma \sim \sum_{k=1}^K \pi_{sk} \phi(\cdot; \theta_k^* - \mu_s, \sigma), \quad \mu_s = \sum_k \pi_{sk} \theta_k^*.$$

The shared atoms let evidence about the residual *shape* flow from the larger source to the smaller; the per-source weights and the centering shifts μ_s retain source-specific flexibility.

6. HDP_σ (HDP + per-source scale). Layer the per-source scale of iDP_σ onto the HDP of HDP: shared atoms θ_k^* , per-source weights π_{sk} and centering shifts μ_s , *and* per-source scales σ_s . This is the prior the methodology section adopts.

At-a-glance summary and notation key. Table ?? arranges the six priors along the shape / stratification / borrowing axis (rows) and the scale-sharing axis (columns). The math symbols in the second and fourth columns are the abbreviations used throughout the remainder of the document; the third and fifth columns give the corresponding value of the `error_dist` argument of `FusionForest`, for use when running the code.

1.4 Data-generating process

We generate observational and randomised data with covariates $X \in [0, 1]^5$, treatment $A \in \{0, 1\}$, source indicator $S \in \{0, 1\}$ (1 = RCT, 0 = RWD), and continuous response $Y = m_0(X) + A\tau(X) + (1 - S)U + \varepsilon$, where:

$$m_0(X) = 2X_1 - X_3^2,$$

$$\tau(X) = 1.5 + 2X_1,$$

$$U \sim N(0, 1) \text{ in the RWD only (unmeasured confounder),}$$

$$A | S = 1 \sim \text{Bernoulli}(0.5),$$

$$A | S = 0 \sim \text{Bernoulli}(\sigma_{\text{logit}}(0.8U)).$$

The residual ε depends on the source S and the scenario.

Scenario A: SHARED. $\varepsilon | S = s \sim N(0, 1)$ for both sources. Null case; all priors should perform similarly.

Scenario B: SCALE. $\varepsilon | S = 1 \sim N(0, 0.5^2)$, $\varepsilon | S = 0 \sim N(0, 1.5^2)$. Sources differ in spread only.

Scenario C: SHAPE + SCALE. For each source s , ε is a two-component Gaussian location mixture on atoms $\{-1.5, +1.0\}$ with per-source weights $\pi_{\text{rct}} = (0.20, 0.80)$, $\pi_{\text{rwd}} = (0.75, 0.25)$, recentered to mean zero per source, plus a residual Gaussian with $\sigma_{\text{rct}} = 0.5$, $\sigma_{\text{rwd}} = 1.5$. Sources share the atom *support* but differ in both weights and scale.

Scenario D: TAIL SHARING. Atoms $\{-3.0, +0.5\}$, per-source weights $\pi_{\text{rct}} = (0.05, 0.95)$, $\pi_{\text{rwd}} = (0.40, 0.60)$, recentered to mean zero per source, with $\sigma_{\text{rct}} = 0.4$ and $\sigma_{\text{rwd}} = 0.8$. The RCT alone barely sees the left-tail atom—in expectation $0.05 n_{\text{rct}}$ observations. This is the regime where HDP atom-borrowing should pay off, because the RWD informs the RCT about the location of a residual feature the RCT would otherwise be too small to identify.

1.5 Sample-size scan

Each scenario is run at three RCT sample sizes, $n_{\text{rct}} \in \{50, 150, 300\}$, with $n_{\text{rwd}} = 300$ fixed throughout. We expect the relative advantage of HDP_σ over iDP_σ (its closest non-borrowing competitor) to grow as n_{rct} shrinks, especially in scenarios C and D.

1.6 MCMC and reproducibility

All forests use $M = 200$ trees for the prognostic component, $M = 100$ for the treatment effect, and $M = 200$ for the deconfounding component, with the default priors of **FusionForest**. The DP truncation level is $K = 50$ and the atom-prior scale is $\sigma_\theta = 0.5$ on the standardised response scale (defaults). Each chain runs 5000 burn-in iterations followed by 5000 post-burn-in draws. Per-cell results are averaged over M replications, with M matched to the number of available cores so that each replication runs on a dedicated core; each replication fits all six priors serially. We target $M \geq 100$ replications per cell.

The full simulation script is at `examples/simulation_error_dist_hpc.R`; the entire posterior state needed to compute every metric in Section ?? is stored in a single tidy data frame and saved to `simulation_error_dist_hpc_output.rds`.

1.7 Hypotheses and reporting plan

We will report results as four blocks of tables and figures, mirroring the four metric families.

H1 (overall CATE). HDP_σ matches the best of the alternatives on RMSE_τ in scenario A (no truth-misspecification regime), strictly outperforms in scenarios B–D, and never loses coverage.

Table 2: CATE RMSE by scenario, RCT sample size, and residual prior. Lowest value per row in bold. See Table ?? for the prior abbreviations.

Scen.	n_{rct}	G	sDP	iDP	iDP $_{\sigma}$	HDP	HDP $_{\sigma}$
A	50	0.471	0.473	0.470	0.476	0.469	0.466
A	150	0.356	0.357	0.355	0.354	0.355	0.354
A	300	0.290	0.290	0.291	0.291	0.291	0.292
B	50	0.495	0.494	0.412	0.388	0.409	0.383
B	150	0.338	0.317	0.261	0.256	0.266	0.256
B	300	0.253	0.224	0.210	0.210	0.214	0.211
C	50	0.571	0.574	0.538	0.526	0.538	0.520
C	150	0.413	0.410	0.380	0.329	0.388	0.353
C	300	0.332	0.272	0.230	0.223	0.269	0.224
D	50	0.561	0.568	0.426	0.445	0.422	0.431
D	150	0.393	0.350	0.238	0.228	0.277	0.240
D	300	0.308	0.220	0.186	0.184	0.200	0.186

H2 (sample-size scan). In scenarios C and D, the gap between HDP $_{\sigma}$ and iDP $_{\sigma}$ grows as n_{rct} shrinks; in scenario D at $n_{\text{rct}} = 50$ the gap is largest and the RCT-only residual sd of iDP $_{\sigma}$ is biased toward the tighter Gaussian centre, while HDP $_{\sigma}$ recovers the tail via the RWD-informed shared atom.

H3 (mixture parsimony). K_s^{eff} and the number of occupied components are smaller for HDP priors than for the independent DPs in every cell, reflecting the shrinkage induced by the top-level γ -prior.

H4 (identifiability check). $|\mu_0 - \mu_1|$ and $\max_k |\theta_k^*|$ remain bounded on the scale implied by the DGP and the atom prior. Where they drift larger, this will be documented but does not affect CATE conclusions, because the posterior of $\theta_k^* - \mu_s$ remains tight.

1.8 Results

We ran $M = 191$ replications per cell on a 192-core node (one core held in reserve). All tables below report the mean across replications.

CATE estimation

Table ?? reports the RMSE of $\hat{\tau}$ across the twelve (scenario, n_{rct}) cells. The headline result is that the scale-aware priors (iDP $_{\sigma}$ and HDP $_{\sigma}$) match each other on CATE in every cell and dominate the equal- σ priors as soon as the truth has unequal source spreads (scenarios B–D), with relative RMSE reductions of 25–40%. The advantage of HDP atom-sharing over independent per-source DPs (iDP $_{\sigma}$ versus HDP $_{\sigma}$) is essentially zero on CATE, but neither is HDP penalised: it matches the best in every cell. Coverage of the nominal 95% credible intervals is high and stable across modes (≥ 0.98 throughout); intervals are correspondingly narrower for the scale-aware modes than for G.

Table 3: Total residual-sd error $\hat{\sigma}_s^{\text{tot}} - \text{sd}(\varepsilon_s^{\text{true}})$ at $n_{\text{rct}} = 150$. Negative values: under-estimating the residual spread. Prior abbreviations as in Table ??.

Mode	Scen. A		Scen. B		Scen. C		Scen. D	
	RWD	RCT	RWD	RCT	RWD	RCT	RWD	RCT
G	−0.45	−0.46	−0.89	0.11	−1.15	−0.42	−1.20	−0.16
sDP	−0.20	−0.21	−0.47	0.53	−0.72	0.01	−0.56	0.48
iDP	−0.06	−0.30	−0.03	−0.16	−0.27	−0.40	−0.04	−0.09
iDP $_{\sigma}$	0.25	0.05	0.17	0.11	0.13	0.06	0.10	0.09
HDP	−0.25	−0.34	−0.02	−0.02	−0.37	−0.40	−0.04	−0.02
HDP $_{\sigma}$	0.25	0.07	0.18	0.18	0.14	0.11	0.15	0.20

Table 4: Effective number of mixture components per source, $K_s^{\text{eff}} = 1/\sum_k \pi_{sk}^2$, at $n_{\text{rct}} = 150$. HDP atom-sharing yields a 3–5 $\times$ reduction.

Mode	Scen. A		Scen. B		Scen. C		Scen. D	
	RWD	RCT	RWD	RCT	RWD	RCT	RWD	RCT
iDP	13.1	7.0	17.0	1.5	15.0	3.8	11.0	1.7
iDP $_{\sigma}$	10.5	6.9	12.2	1.5	12.1	3.8	5.7	1.7
HDP	3.8	2.2	8.1	1.0	5.9	1.3	4.7	1.1
HDP $_{\sigma}$	2.4	2.1	2.6	1.1	2.7	1.7	2.2	1.2

Per-source residual-distribution recovery

Table ?? reports the apples-to-apples total residual sd error at $n_{\text{rct}} = 150$. The equal- σ priors (G, sDP) systematically under-estimate the wider RWD source by 0.5–1.2 across B–D—a substantively large miscalibration. The two scale-aware priors recover both per-source residual sds within roughly 0.2. Kolmogorov–Smirnov distances on the per-source residual CDFs (not shown) cluster tightly across modes ($\text{KS}_{\text{rwd}} \approx 0.05$ – 0.07 , $\text{KS}_{\text{rct}} \approx 0.07$ – 0.13) with the scale-aware priors typically a touch lower on the RCT side, but the gap is small.

Mixture parsimony

This is the clearest qualitative win for the HDP priors. Table ?? shows the effective number of mixture components per source. HDP atom sharing reduces K_s^{eff} by a factor of three to five relative to the independent DPs, and HDP $_{\sigma}$ is the sparsest overall (typically 2–3 effective components per source). The number of jointly-occupied components n_{shared} (not shown) is also substantially smaller under HDP, reflecting genuine sharing rather than coincidence.

HDP identifiability check

Table ?? reports the posterior means of $|\mu_0 - \mu_1|$ and $\max_k |\theta_k^*|$ across all twelve cells for the two HDP priors. Both quantities drift to magnitudes of 10 and beyond—far above the prior atom scale $\sigma_{\theta} = 0.5$ on the standardised response scale and the true DGP-implied mean shifts (zero in A, of order one in C, of order 1.5 in D). This is the μ_s / θ_k^* non-identifiability in action: the likelihood is invariant under $\theta_k^* \mapsto \theta_k^* + c$, $\mu_s \mapsto \mu_s + c$, and the sampler drifts along that manifold. Crucially, this drift does *not* compromise the quantities that matter for inference: CATE RMSE (Table ??) is identical to that of iDP $_{\sigma}$, total residual sds (Table ??) are recovered correctly, and the mixture diagnostics in Table ?? are well behaved.

Table 5: HDP identifiability diagnostics: posterior means of $|\mu_0 - \mu_1|$ and $\max_k |\theta_k^*|$, on the original response scale, for the two HDP priors. Both drift well above what the DGP and the atom-prior scale ($\sigma_\theta = 0.5$ standardised) would imply; CATE inference is unaffected.

Scen.	n_{rct}	HDP		HDP $_\sigma$	
		$ \mu_0 - \mu_1 $	$\max_k \theta_k^* $	$ \mu_0 - \mu_1 $	$\max_k \theta_k^* $
A	50	12.3	3.8	11.4	4.0
A	150	14.6	5.6	13.4	5.9
A	300	15.6	7.3	14.5	7.2
B	50	13.9	3.7	10.2	3.9
B	150	16.0	7.6	14.3	6.3
B	300	17.0	13.1	15.6	7.5
C	50	13.1	4.1	11.4	4.4
C	150	15.7	7.0	14.5	7.0
C	300	17.1	9.6	14.7	8.4
D	50	12.3	4.8	11.5	4.5
D	150	16.1	9.3	14.7	8.3
D	300	17.4	12.1	15.5	9.1

Revisiting the hypotheses

- **H1** (HDP $_\sigma$ is the strict CATE winner): *partially supported*. It is never beaten on CATE, matches the best in 11/12 cells, but ties rather than strictly dominates iDP $_\sigma$. The two scale-aware priors are statistically indistinguishable on CATE in these scenarios.
- **H2** (HDP–DP gap grows as n_{rct} shrinks): *not supported in scenario D*. At $n_{\text{rct}} = 50$ plain HDP (no per-source scale) is the best individual mode in scenario D, suggesting that with very few RCT observations the extra σ_s parameter is harder to identify than the shared atoms it borrows. In scenarios B and C the gap between scale-aware modes and the rest does shrink with sample size, as expected.
- **H3** (mixture parsimony): *fully supported*. HDP priors use 3–5 $\times$ fewer effective components than independent DPs in every cell.
- **H4** (identifiability bounded): *not supported, but with no inferential cost*. $|\mu_0 - \mu_1|$ and $\max_k |\theta_k^*|$ drift well above the prior scale, but CATE estimates and total residual sds are unaffected.

Headline takeaway. HDP $_\sigma$ is the safe default among the six priors: it never loses on CATE, it recovers per-source residual scales correctly, and it does so with substantially fewer effective mixture components than the alternatives. Its main practical cost is the μ_s / θ_k^* non-identifiability, which must be addressed in reporting (see below) but does not affect inference on the estimands of interest.

Comment on identifiability (for the discussion). The HDP construction identifies only the centered locations $\theta_k^* - \mu_s$. Posterior summaries that condition on a specific component label—in particular the unconstrained atom locations θ_k^* and the absolute means μ_s —are therefore not intrinsically meaningful and can drift along the manifold $\theta_k^* \mapsto \theta_k^* + c$, $\mu_s \mapsto \mu_s + c$ without changing the likelihood. The CATE estimand τ is unaffected: it depends on the conditional mean only through the centering constraint $\mathbb{E}[\varepsilon | S = s] = 0$. We recommend reporting and plotting only quantities invariant to this drift—in particular the per-source residual densities $\hat{f}_s(w) = \sum_k \pi_{sk} \phi((w - \theta_k^* + \mu_s)/\sigma_s) / \sigma_s$ and the centered atoms $\theta_k^* - \mu_s$.

2 Simulation on the prior specification

[Historical note on the parametrisation in this section] The Phase-2 grid below was conducted under an earlier variance-budget parametrisation with per-forest shrinkage factors (k_d, k_τ, k_c) in a *tightness* convention (larger $k \Rightarrow$ tighter prior). The current methodology has since dropped this framework in favour of the uniform Chipman calibration $\sigma_{h,f} = k_f/(2\sqrt{m_f})$ with $k_f = 1$ across all forests; the present section is retained as the record of how that default was arrived at. The five Phase-2 configurations (Default, Efficiency, Robust, Equal-share, τ -deep) and the associated (k_d, k_τ, k_c) tuples should be read as the experimental conditions of the simulation, not as the methodology default.

2.1 Goal

The methodology section introduces a variance-budget calibration of the four BART forests $(m_0^{\text{sh}}, d, \tau, c)$, in which a single budget ω^2 is split among the forests through per-forest shrinkage scales k_f . This simulation has two aims.

1. **Pick a default** (k_d, k_τ, k_c) . Among five candidate variance-budget allocations, identify the one that delivers the lowest CATE RMSE at nominal coverage across a grid of confounding scenarios.
2. **Probe the bias–efficiency trade-off.** When the real-world data are confounded, a prior that is too permissive on c leaks bias into τ ; a prior that is too tight forfeits efficiency. The grid spans both regimes so that the chosen default can be defended as the Pareto-frontier choice rather than as a per-scenario winner.

Tree counts and depth-prior parameters are held at the methodology defaults in this study; a one-at-a-time sensitivity sweep around the chosen default is deferred to a follow-up.

2.2 Estimands and metrics

The only estimand scored is the CATE on the log-time scale, $\tau(x) = \mathbb{E}[\log T(1) - \log T(0) \mid X = x]$, evaluated *in-sample* on the combined training data ($n = n_1 + n_0$ points per replication: RCT plus RWD). Bayesian shrinkage from the BART prior plays the role that a hold-out set would play under maximum-likelihood inference: the model is a regularised estimator of τ , and the in-sample posterior mean is exactly the quantity returned to a downstream analyst, so in-sample evaluation is the most direct operational test. For each replicate we record the posterior mean and the equal-tailed 95% credible interval of τ at each training point and compute four summaries, averaged over replications.

- *RMSE.* $\text{RMSE}_\tau = (n^{-1} \sum_{i=1}^n (\hat{\tau}(X_i) - \tau(X_i))^2)^{1/2}$.
- *Pointwise 95% coverage.* $n^{-1} \sum_{i=1}^n \mathbf{1}\{\tau(X_i) \in \text{CI}_{95}(X_i)\}$; nominal target 0.95.
- *Mean credible-interval width.* An efficiency proxy.
- *Bias.* $n^{-1} \sum_{i=1}^n (\hat{\tau}(X_i) - \tau(X_i))$ (signed); detects confounding leakage.

The confounding function c is an internal nuisance and is not scored directly.

Table 6: The five prior configurations compared in this simulation. The columns k_d, k_τ, k_c are in the tightness convention (larger $\Rightarrow$ tighter); $k_{\text{sh}} \equiv 1$ throughout. Trees are listed in the order $(m_{\text{sh}}, m_d, m_\tau, m_c)$. Depth columns are (base, power).

Config	k_d	k_τ	k_c	trees	depth τ / c	role
Default	2.5	1.7	4	200, 50, 200, 200	(0.95, 3) / (0.25, 3)	proposed
Efficiency	4	1.7	10	200, 50, 200, 200	(0.95, 3) / (0.25, 3)	trust RWD
Robust	2	1.7	2	200, 50, 200, 200	(0.95, 3) / (0.25, 3)	discount RWD
Equal-share	1	1	1	200, 50, 200, 200	(0.95, 3) / (0.25, 3)	agnostic baseline
τ -deep	2.5	1.7	4	200, 50, 200, 200	(0.95, 2) / (0.25, 3)	richer HTE

2.3 Compared prior configurations

The leaf prior of each forest is $h_{j,a} \sim N(0, \omega_f^2)$ per leaf, with $\omega_f = k_f^{\text{code}} / \sqrt{m_f}$. The five configurations differ in their setup-side (k_d, k_τ, k_c) , in the *tightness* convention (larger $k \Rightarrow$ tighter prior). The setup-side values are mapped to the code-side leaf-prior scales by

$$s_0 = (1 + k_d^{-2} + k_\tau^{-2} + k_c^{-2})^{-1/2}, \quad k_{\text{sh}}^{\text{code}} = s_0, \quad k_f^{\text{code}} = s_0 / k_f \quad (f \in \{d, \tau, c\}),$$

which fixes the total prior variance $\text{Var}_\pi(m_0^{\text{sh}}(X)) + \text{Var}_\pi(d(X)) + \text{Var}_\pi(\tau(X)) + \text{Var}_\pi(c(X)) = 1$ on the standardised log-time scale. The depth columns in Table ?? are (base, power), i.e. the non-terminal probability base $(1 + d)^{-\text{power}}$ of ?; the baseline and deviation forests use the BART default (0.95, 2) throughout.

The Default configuration encodes a BCF-style ordering with the largest variance on the baseline, a moderate budget on τ , a tighter prior on the RWD deviation d , and the tightest prior on the confounding function c . The Default depth on τ is (0.95, 3) to enforce shallow trees (?); on c the base parameter is dropped to 0.25 to enforce near-constant c trees, motivated by the structural argument that $c \equiv 0$ corresponds to no unmeasured confounding in the RWD. Efficiency tightens the priors on d and c to lean on the RWD; Robust loosens them; Equal-share is the agnostic 25% split per forest and is included to demonstrate that the BCF-style ordering is empirically justified; τ -deep relaxes the τ depth penalty to the BART default to allow richer heterogeneity.

2.4 Data-generating process

We generate observational and randomised data with covariates $X \in [0, 1]^{10}$, treatment $A \in \{0, 1\}$, source indicator $S \in \{0, 1\}$ (1 = RCT, 0 = RWD), and continuous log-survival response

$$\log T = f_0(X, U) + A \tau^*(X, U) + \sigma \varepsilon, \quad \varepsilon \sim N(0, 1).$$

Five covariates $X_1, \dots, X_5$ are active and five $X_6, \dots, X_{10}$ are noise; all are drawn iid $U(0, 1)$ in both sources, with no covariate-shift between RCT and RWD. Let $Z_j = X_j - 0.5$. The structural components are

$$\begin{aligned} f_0(X, U) &= 0.7 \sin(\pi Z_1) + Z_2 Z_3 + (Z_4^2 - \tfrac{1}{12}) + \beta_U U, \\ \tau^*(X, U) &= \tau(X) + \gamma_U U, \\ \tau(X) &= \tau_0 + \tau_1 Z_1 + \tau_2 \mathbf{1}\{X_2 > 0.5\}, \end{aligned}$$

with $U \sim N(0, 1)$ independent of X and identically distributed across sources. The unit-level effect τ^* depends on U through γ_U ; since $U \perp\!\!\!\perp X$, the CATE $\tau(x) = \mathbb{E}[\tau^*(X, U) \mid X = x]$ is exactly the function $\tau(X)$ above. Two CATE regimes are studied: the *homogeneous* regime

$(\tau_0, \tau_1, \tau_2) = (0.5, 0, 0)$ and the *heterogeneous* regime $(0.5, 0.4, 0.3)$. The noise scale is $\sigma = 1$, giving a signal-to-noise ratio of approximately one on the standardised log-time scale.

Treatment in the RCT is randomised, $A \mid S = 1 \sim \text{Bernoulli}(0.5)$; in the RWD treatment depends on observed and unmeasured covariates,

$$A \mid S = 0 \sim \text{Bernoulli}(\text{expit}(\alpha_0 + \alpha_X(Z_1 + Z_2) + \alpha_U U)),$$

with $\alpha_X = 1$ and α_0 tuned per scenario so that the marginal treatment prevalence $\Pr(A = 1 \mid S = 0) \approx 0.5$. The parameter α_U governs the strength of unmeasured confounding through the U - A association.

Confounding scenarios. Four scenarios span the $(\alpha_U, \beta_U, \gamma_U)$ space:

- **S0 (no confounding):** $(0, 1, 0)$. $U \perp\!\!\!\perp A \mid X, S = 0$, so the implied $c(x) \equiv 0$ and the RWD is unconfounded.
- **S1 (mild):** $(0.5, 1, 0)$. Small prognostic-imbalance term in c .
- **S2 (strong):** $(1.5, 1.5, 0)$. Large prognostic-imbalance term.
- **S3 (effect-modification):** $(1, 1, 0.5)$. Moderate c with an added effect-modification component $\gamma_U \mathbb{E}[U \mid x, A = 1, S = 0]$.

Censored and uncensored variants. The simulation is run in two variants. In the *censored variant* an administrative threshold c^* is applied to every observation, with c^* tuned once at script start (via a large pre-simulation) so that the expected censoring rate equals 35% under the central cell (S1, heterogeneous, largest sample size); the same c^* is then used across all cells. In the *uncensored variant* the threshold is removed ($\delta_i \equiv 1$) and $\log T$ is standardised to unit variance via the sample standard deviation before fitting. The uncensored variant serves as a clean reference that removes the wobble introduced by the preliminary AFT-based scale estimate that the censored fit relies on.

2.5 Sample sizes and grid

The full primary grid crosses four confounding scenarios, two CATE regimes (homogeneous, heterogeneous), and two sample-size pairs $(n_1, n_0) \in \{(250, 1000), (100, 400)\}$, yielding $4 \times 2 \times 2 = 16$ cells. Within each cell the five prior configurations of Table ?? are fitted on the same simulated dataset (common random numbers across configurations) so that performance differences are not inflated by dataset-to-dataset noise.

2.6 MCMC and reproducibility

All forests use the tree counts in Table ?. The residual is Gaussian (`error_dist = "gaussian"`) and treatment is centred (`treatment_coding = "centered"`). Each fit runs a *single* MCMC chain of 2000 warm-up plus 2000 retained iterations, with no thinning; standardisation of $\log T$ is handled inside each replicate, on its own data. We run $R = 300$ replications per cell. Replications are parallelised across cores; each replication fits all five configurations serially on its assigned core.

The HPC scripts driving the two variants are `simulations/prior/simulation_censored_hpc.R` and `simulations/prior/simulation_uncensored_hpc.R`. Output is written to `simulation_censored_hpc_out` and `simulation_uncensored_hpc_output.rds` respectively, in long form: one row per (scenario, CATE regime, sample-size pair, configuration, replication) cell, carrying the four CATE metrics.

2.7 Hypotheses and reporting plan

H1 (Default sits on the Pareto frontier). Across the four confounding scenarios, the Default configuration achieves the lowest average RMSE_τ subject to maintaining at least 0.90 coverage at every cell.

H2 (Efficiency wins in S0, loses in S2/S3). In S0, where the truth is $c \equiv 0$, the Efficiency configuration ($k_c = 10$) wins on RMSE_τ because the tight prior on c maximises the effective sample size for τ . In S2 and S3, Efficiency shows substantial bias and under-coverage, because the tight prior on c prevents the model from absorbing the RWD confounding.

H3 (Robust is conservative). Robust matches the Default in S2 and S3 but gives a smaller efficiency gain over the agnostic Equal-share baseline in S0 and S1.

H4 (τ -deep helps under heterogeneity, hurts otherwise). In the heterogeneous CATE regime, τ -deep matches or beats the Default on RMSE_τ ; in the homogeneous regime it is slightly worse, because deeper τ trees overfit the constant truth.

H5 (censored vs uncensored). The ranking of configurations is preserved between the censored and uncensored variants. Absolute RMSE_τ is higher under censoring, reflecting the information loss from 35% administrative censoring and the additional uncertainty in the preliminary AFT-based scale estimate.

2.8 Results

We report preliminary results from $R = 190$ replications per cell, $N_{\text{burn}} = N_{\text{post}} = 2000$, on the *uncensored* variant. The full $R = 300$ runs and the censored variant will replace these numbers in a later revision; the relative ordering of the configurations is expected to be stable. All tables in this subsection average over the 190 replications.

The homogeneous and heterogeneous CATE regimes produce essentially identical relative rankings of the five configurations. We therefore present the heterogeneous CATE as the primary result and report the homogeneous CATE for completeness in Table ??.

CATE RMSE

Table ?? reports RMSE_τ across the eight (scenario, n_1, n_0) cells of the heterogeneous regime. The ranking is consistent across all four scenarios.

Robust is the lowest-RMSE configuration in seven of the eight heterogeneous cells and is beaten only in S0 by **Efficiency** (and even there only by 0.01–0.03). Across S1–S3 the gap

Table 7: CATE RMSE by scenario, sample size, and configuration, for the heterogeneous CATE regime. Lowest per row in bold.

Scen.	(n_1, n_0)	Default	Efficiency	Robust	Equal-share	τ -deep
S0	(100, 400)	0.257	0.246	0.276	0.297	0.257
S0	(250, 1000)	0.233	0.225	0.234	0.247	0.235
S1	(100, 400)	0.278	0.363	0.270	0.296	0.278
S1	(250, 1000)	0.241	0.323	0.232	0.249	0.242
S2	(100, 400)	0.572	1.038	0.357	0.377	0.571
S2	(250, 1000)	0.401	0.864	0.298	0.316	0.402
S3	(100, 400)	0.413	0.680	0.310	0.337	0.412
S3	(250, 1000)	0.318	0.585	0.269	0.287	0.319

Table 8: CATE RMSE by scenario, sample size, and configuration, for the homogeneous CATE regime. Lowest per row in bold.

Scen.	(n_1, n_0)	Default	Efficiency	Robust	Equal-share	τ -deep
S0	(100, 400)	0.224	0.215	0.236	0.266	0.225
S0	(250, 1000)	0.214	0.205	0.211	0.229	0.216
S1	(100, 400)	0.250	0.344	0.236	0.273	0.251
S1	(250, 1000)	0.223	0.313	0.204	0.224	0.224
S2	(100, 400)	0.583	1.038	0.328	0.342	0.581
S2	(250, 1000)	0.407	0.885	0.284	0.303	0.407
S3	(100, 400)	0.414	0.690	0.277	0.306	0.414
S3	(250, 1000)	0.308	0.597	0.240	0.260	0.309

to the runners-up grows: Robust beats Default by 25–40% on RMSE in S2 and 25–30% in S3. Equal-share is the second-best configuration in the confounded scenarios but consistently 0.03–0.04 above Robust on RMSE. **Efficiency** is the across-the-board worst configuration as soon as confounding is present: its RMSE in S2 is roughly three times that of Robust at both sample sizes. **τ -deep** is indistinguishable from Default in every cell (differences are at the third decimal place), which means the depth-prior change on τ does not materially affect CATE estimation at this signal-to-noise level.

The homogeneous-regime RMSE in Table ?? mirrors this ordering, with absolute values 0.02–0.03 below the heterogeneous regime everywhere.

Bias

Signed bias (Table ??) makes the failure mode of Efficiency explicit. Under no confounding (S0) every configuration is near zero ($|\text{Bias}| \leq 0.10$). Under confounding the bias of Efficiency, Default and τ -deep climbs monotonically with the confounding strength: in S2, Efficiency carries a bias of approximately +1.0 in both heterogeneity regimes (the RWD treated-vs-control contrast is approximately one log-time unit larger than the true CATE). Default and τ -deep carry roughly half that bias. **Robust** and **Equal-share** are the only configurations that keep the bias small under confounding; Equal-share is the smallest-bias option in S2 and S3, with Robust a close second.

The bias pattern explains the RMSE pattern: under confounding the squared bias dominates the

Table 9: CATE bias (signed integrated bias of $\hat{\tau}$) by scenario, sample size, and configuration, for the heterogeneous CATE regime. Smallest $|\text{Bias}|$ per row in bold.

Scen.	(n_1, n_0)	Default	Efficiency	Robust	Equal-share	τ -deep
S0	(100, 400)	−0.057	−0.019	−0.097	−0.077	−0.058
S0	(250, 1000)	−0.036	−0.011	−0.054	−0.042	−0.038
S1	(100, 400)	+0.105	+0.272	−0.031	−0.041	+0.103
S1	(250, 1000)	+0.067	+0.237	−0.015	−0.020	+0.066
S2	(100, 400)	+0.498	+1.000	+0.133	+0.060	+0.495
S2	(250, 1000)	+0.292	+0.824	+0.064	+0.023	+0.290
S3	(100, 400)	+0.303	+0.626	+0.053	+0.008	+0.300
S3	(250, 1000)	+0.181	+0.532	+0.023	+0.000	+0.180

Table 10: Pointwise 95% credible-interval coverage by scenario, sample size, and configuration, for the heterogeneous CATE regime. Nominal target 0.95; configurations under-covering are flagged in italics.

Scen.	(n_1, n_0)	Default	Efficiency	Robust	Equal-share	τ -deep
S0	(100, 400)	0.998	0.999	0.994	0.995	0.999
S0	(250, 1000)	0.996	0.997	0.994	0.996	0.997
S1	(100, 400)	0.997	0.983	0.998	0.997	0.997
S1	(250, 1000)	0.996	0.976	0.996	0.996	0.997
S2	(100, 400)	0.945	<i>0.512</i>	0.995	0.996	0.953
S2	(250, 1000)	0.966	<i>0.535</i>	0.991	0.994	0.971
S3	(100, 400)	0.982	<i>0.835</i>	0.994	0.994	0.984
S3	(250, 1000)	0.986	<i>0.821</i>	0.995	0.996	0.989

squared variance for every configuration except Robust and Equal-share, so the configurations that fail to absorb the RWD bias into c pay for it in RMSE.

Coverage

Table ?? reports pointwise 95% credible-interval coverage. All configurations except Efficiency exceed the nominal 0.95 target everywhere (in fact they over-cover at ≥ 0.94 in every cell). The variance-budget calibration is therefore generally conservative; intervals are wider than strictly needed but the nominal level is not violated. **Efficiency** is the only configuration that under-covers, and severely so: 0.51–0.54 in S2 and 0.81–0.84 in S3.

Mean credible-interval widths (not tabulated) lie between 1.35 and 2.25 across all cells and configurations and differ by less than 10% between any two configurations within a cell. Efficiency does *not* buy meaningful interval-width reduction over Robust, even though it shrinks c harder; the widths are nearly the same, so the extra coverage loss in S2/S3 comes from a shift in the posterior centre rather than from artificially tight intervals.

Revisiting the hypotheses

- **H1** (Default is Pareto-optimal): *not supported*. Default is dominated by Robust in S1–S3 on both RMSE and bias and is essentially tied with Robust in S0. The choice of $k_c = 4$ in

Default over-tightens the confounding prior; the data prefer the looser $k_c = 2$ of Robust.

- **H2** (Efficiency wins S0, loses S2/S3): *fully supported*. Efficiency has the lowest RMSE in S0 by a small margin and the largest RMSE and largest bias in S1, S2 and S3 with a coverage of only 0.51–0.54 in S2.
- **H3** (Robust is the conservative choice): *partially supported*. Robust is conservative in coverage (≥ 0.99 everywhere) but is also the lowest-RMSE configuration in seven of eight heterogeneous cells. “Conservative” over-states its cost.
- **H4** (τ -deep helps under heterogeneity): *not supported*. τ -deep is indistinguishable from Default at three decimal places in every cell, in both CATE regimes. The depth-prior change on τ has no practical effect at this signal-to-noise level.
- **H5** (censored vs uncensored ranking preserved): *to be checked* once the censored run completes; the ranking above is from the uncensored run only.

Headline takeaway (Phase 2). On this $R = 190$ preliminary run, **Robust** is the across-the-board winner among the five Phase-2 configurations: lowest CATE RMSE in S1–S3, near-best in S0, no coverage failures anywhere, and the smallest bias outside of Equal-share (which is itself competitive). The “Default” configuration $(k_d, k_\tau, k_c) = (2.5, 1.7, 4)$ is too tight on c ; the data prefer the looser $k_c = 2$ of Robust. Within the variance-budget framework this would motivate adopting Robust $(k_d, k_\tau, k_c) = (2, 1.7, 2)$ as the new default.

Caveat on coverage and the follow-up sweeps. All non-Efficiency configurations over-cover (≥ 0.94 everywhere). This indicates the variance-budget calibration is too generous: the budget $\omega^2 = 1$ on standardised $\log T$ allocates more prior variance to the structural component than is coherent with the inverse- χ^2 residual prior. Phases 2b (v2, $\omega \in \{0.95, 1\}$), 2d (v4, $\omega \in \{0.5, 0.75, 1\}$), and 2e (v5, parametrisation comparison) trace this thread and converge on the uniform Chipman calibration $\sigma_{h,f} = k_f / (2\sqrt{m_f})$ with $k_f = 1$ for every forest as the adopted methodology default. The asymmetric variance-budget allocation diagnosed here gave a small RMSE advantage under strong confounding but at the cost of an opaque calibration; the uniform Chipman default is simpler, matches the BART/BCF defaults of the implementation, and is internally coherent with the residual prior. See the prior-specification subsection of **METHODOLOGY.tex** for the final form.